

[P5] Every r : the residual cover route from $3r - 3$

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Abstract

Sivashankar [2] proved $g(r) \geq 3r - 4$ for every r by deleting the cards of high-degree symbols and bounding the cover number of what remains, a family of maximum degree three, by $4\tau \leq q + r + 4$. He asked whether $+4$ can be replaced by $+3$. Theorem 1 gives a proposed proof that it can, so $g(r) \geq 3r - 3$ for every r . The residual constant $+3$ cannot be lowered [2], and $g(3) = 6$ and $g(4) = 9$ show that $3r - 3$ is attained. Theorem 2 excludes the equality case and gives $g(r) \geq 3r - 2$ for every $r \geq 19$.

Section 4 turns the exclusion of equality into a quantitative statement: when the remainder is large, the loss is linear in r , with a closed form at every remainder size. Section 5 records an exact bookkeeping identity and a covering inequality, and sets out where the route toward a coefficient above 3 stands. No theorem is claimed in Section 5.

The proofs are proposed and have not been independently reviewed. Every finite step is checked exhaustively by a script named in §6. No priority is claimed.

1. Setting

$g(r)$ is the least number of cards in a pairwise intersecting r -uniform family H with $\tau(H) = r$. The known values are $g(3) = 6$, $g(4) = 9$, $g(5) = 13$ [4], and $g(6) = 17$ ([P1]–[P4] of this set). Against $3r$ these are $3r - 3$, $3r - 3$, $3r - 2$ and $3r - 1$. So a bound $g(r) \geq 3r - c$ valid for every r cannot have $c < 3$; a bound $3r - 2$ can hold only from $r = 5$ on, and $3r - 1$ only from $r = 6$ on.

For large r , [2, §4] proves $g(r) \geq ((41 - \sqrt{19})/12 - o(1))r$, about $3.053r$, for r beyond a threshold that is not made explicit. The bounds of this paper have coefficient 3 and are therefore weaker for large enough r . What they add is that they hold, with explicit constants, at every r .

What is borrowed. The peeling reduction, the dual triple system, the maximum matching, the selected witnesses, the first local support bound, and the sharpness family are from [2, §2–3]. They are recalled with proofs so that the argument can be read on its own.

Peeling. Delete the cards containing a symbol of current degree at least four, and repeat until no such symbol is left. If k symbols are chosen and the remainder J has q cards, then

$$|H| \geq q + 4k, \quad r \leq k + \tau(J),$$

because the k chosen symbols together with any cover of J cover H . Hence a bound $4\tau(J) \leq q + r + C$ gives $|H| \geq 3r - C$.

The dual view. For a symbol v of J let B_v be the set of cards containing it. Regard the cards as points. The blocks B_v have size at most three, cover every pair of points, and every point lies in exactly r blocks. A cover of J is a set of blocks meeting every point. For any card E ,

$$q - 1 \leq \sum_{v \in E} (\deg v - 1) \leq 2r, \quad (1.1)$$

so $q \leq 2r + 1$.

Take a maximum matching $M_1, \dots, M_t$ among the distinct three-element blocks. Let C be its union and U the other u points, so $q = 3t + u$; no three-element block lies inside U . A defining symbol of each M_i covers C , and pairing the points of U gives

$$\tau(J) \leq t + \lceil u/2 \rceil. \quad (1.2)$$

For each pair x, y of U choose one symbol w_{xy} in both, the *witness* of the pair. Witnesses are distinct, since a repeated one would lie in four cards or in a triple inside U . Each witness block is $\{x, y\}$ or $\{x, y, z\}$ with $z \in C$.

For each i let F_i be the graph on U whose edge xy is present, with colour z , when the witness block is $\{x, y, z\}$ and $z \in M_i$. The F_i are simple and edge-disjoint. Two disjoint edges of one F_i never have different colours: their blocks would replace M_i by two disjoint triples. For each colour write e for its number of edges and s for its support, and let S_i be the sum of the three supports, $S = \sum_i S_i$.

2. Theorem 1: the sharp residual bound

Theorem 1 (proposed). *Every intersecting r -uniform family J with q cards and maximum degree at most three satisfies $4\tau(J) \leq q + r + 3$.*

Corollary 1. $g(r) \geq 3r - 3$ for every $r \geq 1$.

Proof. Peeling and Theorem 1. $\square$

2.1 Three inequalities

Capacity at U . The witnesses use exactly $u(u - 1)$ incidences on U . The other $ur - u(u - 1)$ incidences each reach at most two cards of C , and all $3tu$ pairs between U and C must be met. Hence

$$3tu - S \leq 2(ur - u(u - 1)). \quad (2.1)$$

Capacity of a colour. Fix $z \in M_i$, and let the edges of colour z in F_i be e in number, with support s . In the card z , the e witnesses cover s points of U , a defining symbol of M_i covers its other two points, and the remaining $q - 3 - s$ points need the other $r - e - 1$ symbols, at most two each. So

$$2e - s \leq 2r - q + 1 =: d. \quad (2.2)$$

Local support ($[2]$ for the first bound).

Lemma 1. *Let F be a simple graph on u vertices with at most three colours, in which disjoint edges always have the same colour. Then $S_F \leq \max(12, u + 4)$. If every colour has $2e - s \leq d$, then $S_F \leq \max(12, u + 2, d + 8)$.*

Proof. Suppose a colour has two disjoint edges, on four vertices W . Every other-colour edge lies in W . If no other colour is present, $S_F \leq u$. If the other-colour edges have no common vertex, every edge of the first colour meets all of them and lies in W too, giving 12. Otherwise at most two other-colour edges exist, sharing a vertex, with support contribution at most four, giving $u + 4$. Fix one of them, g . Every first-colour edge meets g , so $e \geq s - 2$, and $2e - s \leq d$ gives $s \leq d + 4$ and $S_F \leq d + 8$.

If every colour is internally intersecting, each is a star or part of a triangle. With all supports at most four, the sum is at most 12. Otherwise one colour is a star with at least four leaves, every other-colour edge contains its centre, and simplicity gives $S_F \leq u + 2$. $\square$

2.2 The preliminary inequality

Lemma 2 ([2]). $r \geq u + t - 2$.

Proof. Put $\alpha = r - u - t + 2$. If $u = 0$, (1.1) gives $\alpha \geq (r + 5)/3 > 0$. Otherwise (1.1), (2.1) and Lemma 1, with $K = \max(12, u + 4)$, give $t \leq u - 3 + 2\alpha$ and $\alpha \geq 1 - (K - u)t/(2u)$. A negative integer α would give $t \leq u - 5$ and then $\alpha > -1$: for $K = u + 4$ the bound is $-1 + 10/u$, and for $K = 12$ with $u \leq 8$ the needed strict inequality is $u^2 - 13u + 60 > 0$, whose discriminant is negative. $\square$

2.3 The odd equality case

By (1.2) and Lemma 2, if u is even then $4\tau \leq 4t + 2u = q + (t + u) \leq q + r + 2$. If u is odd, $4\tau \leq q + (t + u) + 2$. So Theorem 1 follows once *u odd implies $r \geq u + t - 1$* .

Suppose u is odd and, by Lemma 2, $r = u + t - 2$. Then $d = u - t - 3 \geq 0$, and (2.1) becomes

$$S \geq u(t + 2). \quad (2.3)$$

If $u \geq 11$, (2.3) and Lemma 1 give $t \geq u/2$. Then $d + 8 \leq u + 2$ and $12 \leq u + 2$, so every $S_i \leq u + 2$ and $u(t + 2) \leq t(u + 2)$, that is $u \leq t$, against $t \leq u - 3$.

For odd $u \leq 9$, the conditions $0 \leq t \leq u - 3$ and $u(t + 2) \leq t \max(12, u + 4)$ leave exactly

$$(u, t) \in \{(5, 2), (7, 3), (7, 4), (9, 5), (9, 6)\},$$

with $d = 0, 1, 0, 1, 0$ respectively.

Lemma 3 (small excess). *If every colour has $2e - s \leq 1$ and F has more than one colour, then $S_F \leq 12$, $S_F \neq 11$, and $S_F = 12$ only when F is a properly three-coloured K_4 . If also $u \leq 9$, the same holds when F has one colour.*

Proof. A connected component on $v \geq 2$ vertices has $2e - v \geq v - 2$, so each colour is a matching, or one path on three vertices plus a matching. A colour with a matching of size three leaves the other colours empty, so F has one colour and $S_F \leq u \leq 9$. A colour of support five is P_3 plus an edge, and the other colours are then confined to two edges, giving at most 9. Otherwise every support is at most four. Support four forces two disjoint edges, and two cross-intersecting two-edge matchings are two perfect matchings of one K_4 ; a third colour must be the third. $\square$

The cases $d = 1$. For $(9, 5)$, (2.3) asks $S \geq 63$ and Lemma 3 allows 60. For $(7, 3)$, (2.3) asks 35, so all three S_i are 12: three edge-disjoint copies of K_4 on seven vertices. Two such copies meet in one vertex and leave $K_{3,3}$ plus an isolated vertex, which holds no third K_4 .

The cases $d = 0$. Now every colour is a matching and $q = 2r + 1$, so (1.1) is an equality for every card: every symbol has degree three and any two cards meet once. So every witness block is a triple and the F_i partition the edges of K_u . For $(5, 2)$, $qr = 55$ is not divisible by three. For $u \leq 9$ a local graph has at most six edges: one colour is a matching of at most four edges, and with several colours each is a matching of at most two, since a third disjoint edge would meet no other-colour edge. Six edges occur only as the three perfect matchings of a K_4 , and five only as K_4 minus an edge. For $(9, 6)$ the 36 edges of K_9 would split into six copies of K_4 , needing degree 8 divisible by three. For $(7, 4)$ the 21 edges of K_7 split into four parts of size at most six only as $6 + 6 + 6 + 3$, $6 + 6 + 5 + 4$ or $6 + 5 + 5 + 5$; two edge-disjoint K_4 leave $K_{3,3}$ plus a vertex, which excludes the first two. In the third, after one K_4 on four vertices, each copy of K_4 minus an edge contains a triangle and so takes one of the three edges among the other three vertices; it is then two of the four joined to two of the three, plus that edge. The three pairs among the three are distinct and meet, so edge-disjointness needs three disjoint pairs among four vertices, which is impossible.

This proves Theorem 1. $\square$

2.4 Sharpness, and degree four

For odd n take points $1, \dots, n$ and one symbol for each pair ([2]). Then $q = n$, $r = n - 1$, a cover is an edge cover of K_n , and $\tau = (n + 1)/2$, so $4\tau = q + r + 3$.

Corollary 2. *If J has maximum degree at most four, then $4\tau(J) \leq q + r + 3$.*

Proof. Take a maximal matching of a four-element blocks; their defining symbols cover $4a$ cards and the rest has maximum degree three. $\tau \leq a + (q - 4a + r + 3)/4$. $\square$

Peeling at degree five then gives $|H| \geq (15r - q - 15)/4$ for the size q of the remainder, which improves the additive term in the corresponding step of [2] but not its coefficient.

3. Theorem 2: excluding equality

Theorem 2 (proposed). $g(r) \geq 3r - 2$ for every $r \geq 19$.

Suppose $|H| = 3r - 3$ and $\tau(H) = r$ with $r \geq 5$. We derive $r \leq 18$.

3.1 The equality parameters

If H had a symbol of degree at least five, peeling it first would give $|H| \geq |J| + 4(r - \tau(J)) + 1 \geq 3r - 2$ by Theorem 1. If every degree were at most three, $3r - 3 \leq 2r + 1$ would force $r \leq 4$. So the maximum degree is four. (At $r = 3$ and $r = 4$ the bound $3r - 3$ is attained, and this is where the argument stops.)

Take a maximum matching of a four-element blocks, delete its $4a$ cards, and let J be the rest, of maximum degree three and $q = 3r - 3 - 4a$ cards. The a defining symbols and a cover of J cover H , so $\tau(J) \geq r - a$, while Theorem 1 gives $4\tau(J) \leq q + r + 3 = 4(r - a)$. So Theorem 1 is an equality for J , which by §2.3 forces u odd, $\tau(J) = r - a$, $r = u + t - 1$ and $q = 3t + u$. Consequently

$$u = 2a + 3, \quad t = r - 2a - 2, \quad d = u - t - 1 \geq 0, \quad r = 2u - 2 - d, \quad (3.1)$$

and (2.1) becomes $S \geq ut$. Write $H_0 = (u - 1)/2$.

3.2 Pointwise capacity

Lemma 4. *For $x \in U$ let S_x be the number of cards of C met by the witnesses at x . Then $S_x \geq t$. In one F_i , if x carries E edges in k colours, then $E - k \leq d$.*

Proof. The card x has exactly $u - 1$ witnesses, so exactly $r - (u - 1) = t$ other symbols. They meet at most $2t$ cards of C , and all $3t$ must be met, so $3t - S_x \leq 2t$. The $u - 1$ witnesses at x reach S_x distinct cards of C , so the number of witnesses “wasted” at x is $u - 1 - S_x \leq u - 1 - t = d$. Edges of one colour at x reach one card, so E edges in k colours waste $E - k$. $\square$

Lemma 5. *In the equality setting every F_i with more than one colour has $S_i \leq \max(12, d + 8)$ and $S_i - e(F_i) \leq 6$. A one-coloured F_i has $S_i - e(F_i) \leq H_0$.*

Proof. The first case of Lemma 1 already gives $\max(12, d + 8)$. In the star case, F_i is a star at one vertex with E edges in $k \leq 3$ colours, and $S_i = E + k \leq d + 2k \leq d + 6$ by Lemma 4. For the second bound: with two disjoint edges of one colour on W , either every support lies in W and each colour has $s - e \leq 2$, or at most two other-colour edges exist, contributing at most two, while the first colour has $e \geq s - 2$. Otherwise every colour is a star or triangle with $s - e \leq 1$. A one-coloured graph has $s - e$ at most its number of components, at most H_0 . $\square$

3.3 Two augmentation bounds

Call a one-coloured F_i *large* if its matching number is at least four, and let m be their number.

Lemma 6. $2m \leq t$.

Proof. A large F_i uses one colour z_i ; its other two points of M_i are met by no witness. So every $x \in U$ must meet those $2m$ points with its t non-witness symbols. No such symbol meets two of them. Within one M_i , its triple and an edge of F_i avoiding x would replace M_i by two triples. Across M_i and M_j , an edge of F_i avoiding x and an edge of F_j avoiding x and those two endpoints, which matching number four provides, give three triples in place of two. Either way the matching was not maximum. $\square$

Lemma 7. $u \leq \max(12, d + 8)$.

Proof. Suppose not. By Lemma 5 every F_i with more than one colour has $S_i < u$, and a one-coloured F_i has $S_i \leq u$. Since $S \geq ut$, every F_i is one-coloured and spans U . Then $2e - u \leq d$ gives $e \leq (u + d)/2$, and its matching number is at least $u - e \geq (u - d)/2 > 4$. So $m = t$, and Lemma 6 forces $t = 0$, whereas $t = u - d - 1 \geq 8$. $\square$

3.4 The deleted cards

For each of the $4a$ deleted cards A let L_A be the graph on U whose edge xy is present when $w_{xy} \in A$; write e_A, s_A, ν_A for its edges, support and matching number. Let $T = \sum_i e(F_i)$ and $D = S - T$.

A witness of degree two in J lies in at most two deleted cards, one of degree three in at most one, since degrees in H are at most four; so $\sum_A e_A \leq u(u - 1) - T$. Let $M = \sum_A (u - s_A)$. For $x \in U$ its t non-witness symbols have room for at most $3t$ other cards of H ; they must meet the $3t - S_x$ cards of C not met by witnesses, and every deleted card A whose graph misses x . So at most S_x deleted cards miss x , and $M \leq S$. Using $\nu_A \geq s_A - e_A$ and $4a = 2u - 6$,

$$\sum_A \nu_A \geq u^2 - 5u - D. \quad (3.2)$$

Lemma 8. *Within one group of four deleted cards there are no three disjoint edges from three different graphs L_A .*

Proof. Extend them to a matching of H_0 edges of the complete graph on U , leaving one point x . The witnesses of these edges cover U except x and meet three cards of the group; one symbol of x that lies in the fourth card covers both. With the t defining symbols of C and the $a - 1$ symbols of the other groups this is a cover of H of size $H_0 + 1 + t + a - 1 = r - 1$. $\square$

Lemma 9. *Three graphs each with a matching of size four and a fourth nonempty graph contain three disjoint edges from three different graphs.*

Lemma 10. *The same holds for four graphs with matching numbers at least 3, 3, 4, 4.*

Proof of both. Fix size-four matchings P, Q in two of the graphs and suppose no such triple exists. Call an edge e of another graph *blocked* if no two disjoint edges, one from P and one from Q , avoid it; every edge of the other graphs is blocked. The edges of P and of Q avoiding a blocked edge must cross-intersect. Each side keeps at least two, and one edge cannot meet three disjoint edges, so exactly two survive on each side and they form a four-cycle, which is a component of $P \cup Q$, with both ends of e in both supports. The other graphs supply three disjoint blocked edges (a size-three matching for Lemma 10, a size-four matching for Lemma 9). Their six endpoints lie in both supports, and outside one such four-cycle there are at most four such vertices, so some of them touches it and yields a second four-cycle. So $P \cup Q$ is two disjoint four-cycles on the same eight vertices, and every blocked edge is a diagonal of one of them.

For Lemma 10 the two size-three matchings each consist of three of the four diagonals. The first has both diagonals of one cycle and the second at least one of them; take that one for the second, the other for the first, and an edge of P in the other cycle. For Lemma 9 the three size-four matchings are the three perfect matchings of each of the two copies of K_4 . Take an edge f of the fourth graph. If f removes two vertices of one copy, the edge on its other two vertices and an edge of a different matching in the other copy avoid f . If f removes at most one vertex of each copy, each copy keeps a triangle or all of K_4 , which contains all three matchings, so two edges of different matchings avoid f . In every case a rainbow triple exists. $\square$

Order the matching numbers of one group as $v_1 \leq v_2 \leq v_3 \leq v_4$. Greedy choice gives a rainbow triple when $v_2 \geq 1$, $v_3 \geq 3$ and $v_4 \geq 5$. With Lemmas 8–10, every group has $v_2 = 0$, or $v_3 \leq 2$, or all $v_i \leq 4$ with neither $v_1 \geq 1$ and $v_2 = v_3 = v_4 = 4$ (Lemma 9) nor $v_1, v_2 \geq 3$ and $v_3 = v_4 = 4$ (Lemma 10). So the sum is at most $\max(2H_0, H_0 + 6, 13)$, and with (3.2):

$$D \geq \frac{u^2 - 6u - 3}{2} \quad (u \geq 15), \quad D \geq 39 \quad (u = 13). \quad (3.3)$$

3.5 Finishing

$u \geq 15$. By Lemma 7, $d \geq u - 8$ and $t = u - d - 1 \leq 7$. By Lemma 5, $D \leq 7H_0 = 7(u - 1)/2$. With (3.3) this gives $u^2 - 13u + 4 \leq 0$, false for $u \geq 13$.

$u = 13$. Lemma 7 gives $d \geq 5$. If $d \geq 6$ then $t \leq 6$ and $D \leq 6t \leq 36$, against (3.3). If $d = 5$ then $t = 7$ and every $S_i \leq 13$, so $S \geq 91$ forces every $S_i = 13$. A one-coloured F_i then has $e \leq 9$ and

matching number at least four, so it is large and $S_i - e \leq 6$. A multicoloured F_i cannot be the star case of Lemma 5, which gives at most $d + 6 = 11$, so $S_i = 13 = d + 8$ is the extremal case of Lemma 1: two other-colour edges pa, pb , and a first colour made of ab and six edges at p , so $S_i - e = 13 - 9 = 4$. Lemma 6 gives $m \leq 3$, so $D \leq 6 \cdot 3 + 4 \cdot 4 = 34$, against (3.3).

So $u \leq 11$ and $r = 2u - 2 - d \leq 20$.

$r = 20$. Then $u = 11, d = 0, t = 10, q = 41 = 2r + 1$. As in §2.3 every symbol of J has degree three, but $qr = 820$ is not divisible by three.

$r = 19$. Then $u = 11, d = 1, t = 9$. By Lemma 3 a multicoloured F_i has $S_i \leq 10$ unless it is a K_4 with $S_i = 12$. A one-coloured one has $S_i \leq 11$, and $S_i \leq 7$ unless it is large, since matching number at most three and $2e - s \leq 1$ allow support at most seven. Lemma 6 gives $m \leq 4$. Without a K_4 , $S \leq 11 \cdot 4 + 10 \cdot 5 = 94 < 99 = ut$. So some F_i is a K_4 , on four points W .

For a local graph F_j call (x, c) *eligible* if no edge of colour c meets x and every edge of the other two colours does. Put $\delta_j = 12 - S_j \geq 0$; then $\sum_j \delta_j = 108 - S \leq 9$. A multicoloured F_j has at most three eligible pairs, and any eligible pair forces $S_j \leq 6$, so its count is at most δ_j . A one-coloured F_j of support s has $11 - s = \delta_j - 1$ eligible pairs, or $13 - s = \delta_j + 1$ if it is a path on three vertices; such a path has $\delta_j = 9$, so at most one occurs. The total number of eligible pairs is therefore at most $9 + 1 = 10$.

Now take $x \notin W$; there are seven. No witness at x meets the triple M_i of the K_4 . The card x has nine non-witness symbols and $S_x \leq 10$, so they must meet at least 17 cards of C that the witnesses miss. No symbol joins x to two points of M_i , by augmentation with an edge of the K_4 . So three distinct symbols meet the three points of M_i , and since the other six meet at most 12 new cards, these three reach at least two further new cards $q' \in M_j$. For such a block $\{x, p, q'\}$, an edge f of F_j of another colour avoiding x would give three disjoint triples in place of M_i, M_j : the block itself, f with its colour, and an edge of the four-cycle of the K_4 in the two colours other than p that avoids f . So (x, q') is eligible. That is at least $7 \cdot 2 = 14$ eligible pairs, against 10.

This proves Theorem 2. $\square$

4. A quantitative form: loss linear in r

Drop the equality assumption. For the remainder J of degree-four peeling put

$$\ell = r - u - t + 1, \quad h = t + \ell, \quad d = 2r - q + 1.$$

Every $x \in U$ has h non-witness symbols, and (1.2) gives $B := q + r + 3 - 4\tau(J) \geq \ell$. The argument of Lemma 4 gives $S_x \geq t - 2\ell$, so $S \geq u(t - 2\ell)$, and $u - 1 - S_x \leq d$ still holds, so Lemma 5 holds. With $L = \max(12, d + 8)$, every multicoloured F_i , and every one-coloured F_i that is not large, has $S_i \leq L$. The argument of Lemma 6 gives $2m \leq h$.

Theorem 3 (proposed). *If $u > L$ then*

$$\ell \geq \frac{(u - L)(u - d - 1)}{3u + L}. \quad (4.1)$$

Proof. $u(t - 2\ell) \leq S \leq Lt + (u - L)m \leq Lt + (u - L)(t + \ell)/2$, which rearranges to $(u - L)t \leq (5u - L)\ell$; substitute $t = u - d - 1 + 2\ell$. $\square$

Corollary 3. *If the degree-four peeling of H leaves a remainder with $q \geq 2r - 3$, then $|H| \geq \lceil (28r - 67)/9 \rceil$.*

Proof. Here $d \leq 4$ and $L = 12$. For $u > 12$, (4.1) gives $\ell \geq (u - 21)/3$, and with $r = 2u - d - 2 + 3\ell$ this is $\ell \geq (r - 40)/9$. For $u \leq 12$, $r \leq 22 + 3\ell$ gives the same for $r \geq 13$, and for smaller r the bound is negative. So $B \geq (r - 40)/9$, and $|H| = 3r - 3 + E + B + 4C$ (§5.1) gives the claim. $\square$

The coefficient here is $28/9 \approx 3.111$, but only under the hypothesis on q , which nothing here forces.

The gain at every remainder size. (4.1) holds for every d , not only $d \leq 4$. Writing $q = (2 - \delta)r$ and minimising ℓ over u , the guaranteed gain is, to leading order in r ,

$$\frac{\ell}{r} \geq \frac{(12 + 4\delta) - \sqrt{(12 + 4\delta)^2 - 108(1 - \delta)^2}}{54},$$

which is $1/9$ at $q = 2r$, about 0.048 at $q = 1.75r$, about 0.019 at $q = 1.5r$, about 0.004 at $q = 1.25r$, and 0 at $q = r$. It is positive for every remainder larger than r by a fixed fraction. For remainders smaller than r by a fixed fraction the gain is immediate: $\tau(J) \leq \lceil q/2 \rceil$ gives $B \geq r - q + 1$. What is not covered is a remainder of size close to r , which is where the pair family of §2.4 lives.

5. The route toward a coefficient above 3

No theorem is claimed in this section.

5.1 Bookkeeping

For the degree-four peeling of H , with $n = |H|$, put

$$E = n - q - 4k, \quad B = q + r + 3 - 4\tau(J), \quad C = k + \tau(J) - r.$$

E and C are non-negative, and B is by Theorem 1. Then exactly

$$n = 3r - 3 + E + B + 4C. \quad (5.1)$$

A coefficient above 3 needs $E + B + 4C \geq \varepsilon r - O(1)$ for every H . §4 gives it through B except near $q = r$. Near $q = r$ a lemma about J alone cannot give it, because the pair family has $B = 0$. For the same reason no bound $5\tau \leq q + r + K$ holds for maximum degree four: the pair family of §2.4 on N points forces $K \geq (N + 7)/2$.

5.2 A covering inequality

Every deleted card A meets every card of J , so the symbols of A that occur in J form a cover of J of size at most $r - 1$. If a cover of J of size $\tau(J) + s$, together with a set X of further symbols, meets every card of g deleted groups, then it replaces those groups' symbols in a cover of H , and

$$C \geq g - s - |X|. \quad (5.2)$$

The question near $q = r$ is therefore about J and the covers of J that the deleted cards contain.

5.3 A measurement in the cleanest case

Take J to be exactly the pair family on u points, so that each deleted card contains an edge cover of K_u . A near-perfect matching of K_u is a minimum cover of J . If it meets all four edge covers of a group, that group costs nothing in (5.2).

`code/measure_matching_hits.py` searches for the four edge covers that make this least likely for a random perfect matching. The worst probability found is 0 at $u = 8$ (the configuration of two copies of K_4 that appears in Lemmas 9 and 10), $7/945 \approx 0.0074$ at $u = 10$ (the exhaustive value), about 0.012 at $u = 12$, about 0.017 at $u = 16$ and about 0.020 at $u = 24$. These rise toward $(1 - e^{-1/2})^4 \approx 0.024$, the value for independent events. If a positive lower bound of this kind holds for all u , then (5.2) gives C linear in r in this case. This is a measurement on a model; it is not a proof, and the model leaves out the triples and the non-witness symbols that a real remainder near $q = r$ has.

5.4 Where it stands

Put together, the three cases give a positive ε in principle. As the pieces stand, the weakest point is a window of remainders just above r : there §4 gives only about $\eta^2 r/16$ for $q = (1 + \eta)r$, while the covering argument has to pay for non-witness symbols whose number grows like ηr . The constant that results is far below 0.053 and is not stated as a result. One observation may help: in that window the non-witness symbols at x are mostly blocks $\{x, c, c'\}$ with $c, c' \in C$, and when c and c' lie in the same M_i such a block can replace M_i in the maximum matching at no cost. When they lie in different triples, the same augmentation arguments as in Lemmas 6–10 constrain them.

6. Verification

claim	script
Theorem 1 on every configuration with at most six cards, and at low degree up to nine; the sharpness family	<code>code/verify_residual_bound.py</code>
Lemmas 1, 3, 5 and the $d = 0$ structure of §2.3; the $u = 13, d = 5$ step; the eligible-pair bounds	<code>code/verify_local_graphs.py</code>
Lemmas 9 and 10 on 8, 9 and 10 vertices	<code>code/verify_rainbow_lemmas.py</code> , <code>code/verify_rainbow.c</code>
every arithmetic step of §2.3, §3.5, §4 and §5, and the constant of [2]	<code>code/verify_p5_arithmetic.py</code>
the measurement of §5.3	<code>code/measure_matching_hits.py</code>

The runs are recorded in `results/p5_checks.txt`. These are finite checks of the finite lemmas. The steps that are not finite, namely (2.1), (2.2), Lemma 4, the inequality $M \leq S$, and the augmentation arguments of Lemmas 6 and 8 and of §3.5, have been read but not independently reviewed.

This work was prepared with AI assistance (ChatGPT, OpenAI; Claude, Anthropic), used for algebraic derivation, for drafting and rewriting code and text, for running and re-running the computations, and for auditing the papers against their own scripts. The same tools were used in the review, so the review carries the same caveat. All statements were checked by the author, who is responsible

for them. No priority is claimed and the result has not been independently reviewed by a human or a proof assistant; the repository *README* sets out the division of labour and the open obligations in full.

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